Arun Mathew

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Citizenship: India

Professional Summary

Theoretical and computational astrophysicist with over 10 years of research experience across non-equilibrium plasma astrophysics, radiation–magnetohydrodynamics (MHD), general relativity, cosmology, and higher-order gravity. My work spans the modelling of ionised plasmas in stellar and high-energy astrophysical environments, with a focus on connecting fundamental microphysics to observable phenomena, as well as studies of extended gravity in compact objects and early-Universe physics (including inflation and reheating).

I am the developer of the multi-ion non-equilibrium plasma solver `NEMO` and the non-equilibrium spectral synthesis package `NebulaPy` based in `ChiantiPy` and `NebPy`, enabling self-consistent simulations and predictive modelling of multi-wavelength emission from radio to X-rays. Author of more than 13 refereed publications (8 as first author; h-index 5) in leading journals.

I bring over four years of teaching and mentoring experience across senior secondary, undergraduate, and postgraduate levels, with demonstrated capability in delivering lectures, tutorials, and laboratory sessions in physics and computational methods. I have supervised and supported research projects at undergraduate, master's, and PhD levels, guiding students in developing strong analytical skills, rigorous scientific reasoning, and effective communication, while fostering an inclusive and supportive learning environment that promotes critical thinking and intellectual independence.

Education

Ph.D, Astrophysics and Cosmology — August 2021
Institute: Indian Institute of Technology Guwahati (IITG) — Guwahati, India
Thesis Title: *Generalized uncertainty scenario of quantum gravity in white dwarfs and extended gravity scenarios in quark stars and inflationary cosmology*
Supervisor: Dr. Malay Kumar Nandy

M.S. in Applied Physics — 2013
Institute: Mar Thoma College, Mahatma Gandhi University — Tiruvalla, India

M.S. in Physics (Astrophysics) — 2010
Institute: Indian Institute of Technology Guwahati — Guwahati, India

B.S. in Physics (Major), Mathematics, and Statistics — 2010
Institute: CMS College, Mahatma Gandhi University — Kottayam, India

Postdoctoral Research

Computational and High Energy Astrophysics — 2022-2024
Institute: Dublin Institute for Advanced Studies — Dublin, Ireland
Project: *Synthetic observations and computational modeling of Wolf–Rayet nebulae, investigating their ionization structure and emission properties using MHD simulations*
Mentor: Dr. Jonathan Mackey
Support under the Royal Society Enhanced Research Expenses Grant (RF\ERE\210382)

Non-Academic Experience

Production Operator — March 2025–Present
Syntro Health — Melbourne, Australia
cGMP-compliant manufacturing of clinical trial pharmaceuticals, involving precision processes and rigorous quality and regulatory compliance.

List of Publications

1. **Mathew, A.**, Mackey, J., & Jiménez-Hernández, P., Haworth, T. J., Walch, S., & Mellema, G., "Line emission from non-equilibrium-ionisation simulations of wind-driven bow shocks of hot stars," *Manuscript in preparation*, 2026.

2. Jiménez-Hernández, P., Mackey, J., **Mathew, A.**, et al. "Multiwavelength analysis of the extended nebula around the WO-type Wolf–Rayet star WR 102," *Manuscript in preparation*, 2026.

3. Mackey, J., **Mathew, A.**, Ali, A. A., Haworth, T. J., Brose, R., Green, S., Moutzouri, M., & Walch, S. "Thermal emission from bow shocks III: Variable diffuse X-ray emission from stellar-wind bow shocks driven by dynamical instabilities," *Astronomy & Astrophysics*, 696, A91 (2025). doi:10.1051/0004-6361/202553722

4. Shafeeque, M., **Mathew, A.**, & Nandy, M. K. "Supermassive neutron stars in Starobinsky gravity with causal hybrid stellar matter," *Under review*, 2025. arXiv:2508.09861

5. **Mathew, A.**, Mackey, J., Celeste, M., Haworth, T. J., & Mellema, G. "A multi-ion non-equilibrium solver for ionised astrophysical plasmas with arbitrary elemental abundances," *Astronomy & Astrophysics*, 695, A73 (2025). doi:10.1051/0004-6361/202452373

6. Mackey, J., Jones, T. A. K., Brose, R., Grassitelli, L., Reville, B., **Mathew, A.** "Inverse-Compton cooling of thermal plasma in colliding-wind binaries," *Monthly Notices of the Royal Astronomical Society*, 526, 3099 (2023). doi:10.1093/mnras/stad2839

7. Shafeeque, M., **Mathew, A.**, & Nandy, M. K. "Maximal mass of the neutron star with a deconfined quark core," *Journal of Astrophysics and Astronomy*, 44, 63 (2023). doi:10.1007/s12036-023-09957-5

8. **Mathew, A.** & Nandy, M. K. "Existence of Chandrasekhar's limit in generalized uncertainty white dwarfs," *Royal Society Open Science*, 8, 210301 (2021). doi:10.1098/rsos.210301

9. **Mathew, A.**, Shafeeque, M., & Nandy, M. K. "Stellar structure of quark stars in a modified Starobinsky gravity," *European Physical Journal C*, 80, 615 (2020). doi:10.1140/epjc/s10052-020-8130-4

10. **Mathew, A.** & Nandy, M. K. "Prospect of Chandrasekhar's limit against modified dispersion relations," *General Relativity and Gravitation*, 52, 38 (2020). doi:10.1007/s10714-020-02686-y

11. **Mathew, M.** & Nandy, M. K. "Noncommutative dispersion relation and mass–radius relation of white dwarfs," *Research in Astronomy and Astrophysics*, 18, 151 (2018). doi:10.1088/1674-4527/18/12/151

12. **Mathew, M.** & Nandy, M. K. "Effect of minimal length uncertainty on the mass–radius relation of white dwarfs," *Annals of Physics*, 393, 184 (2018). doi:10.1016/j.aop.2018.04.008

13. **Mathew, M.** & Nandy, M. K. "General relativistic calculations for white dwarfs," *Research in Astronomy and Astrophysics*, 17, 061 (2017). doi:10.1088/1674-4527/17/6/61

Forthcoming Publications

14. **Mathew, A.** & Mackey, J. "Predicting Fe–K emission from the colliding-wind binary WR 140 with non-equilibrium ionisation," *Work in progress*, 2026.

15. **Mathew, A.** & Nandy, M. K. "Primordial reheating in $f(R)$ cosmology by spontaneous decay of scalarons," *To be submitted*, 2025. arXiv:2012.13960

Refereed Conference Proceedings

16. **Mathew, A.** & Mackey, J. "Implementation of multi-ion chemical kinetics for circumstellar nebulae," *Revista Mexicana de Astronomía y Astrofísica Conference Series*, 58, 19 (2024). RMxAC, Vol. 58

17. Mackey, J., Jones, T. A. K., Brose, R., Grassitelli, L., Reville, B., **Mathew, A.** "Performance and scaling of `PION` for modelling colliding-wind binary systems," *Procedia Computer Science*, 240, 82 (2024). doi:10.1016/j.procs.2024.07.011

Competitive Fellowships and Academic Achievements

- Senior Research Fellowship awarded by MHRD, Government of India 2017-2019
- Junior Research Fellowship awarded by MHRD, Government of India 2013-2016
- Graduate Aptitude Test in Engineering (All India Rank: 185) 2013
- Graduate Aptitude Test in Engineering (All India Rank: 827) 2012
- Joint Admission Test for M.Sc. (All India Rank: 189) 2008

Code Design and Development

- `NEMO` – Lead developer of a multi-ion, non-equilibrium ionization solver integrated into `PION`.
- `NebulaPy` – Designed and developed post/pre-processing Python tool to generate synthetic emission maps and line luminosities utilizing the Chianti database.
- Gravity-Matter GR Code – A solver for quark star structures in a perturbative $f(R,T)$ gravity-matter model, using an iterative methods.
- Dynamical Instability Code – Solve stellar structure and dynamical instability analysis of white dwarfs incorporating generalized uncertainty principles equation of state.

High-Performance Computing Experience

- Kay @ ICHEC: Kay has a cluster of 336 nodes where each node has 2x20-core
  Total Allocation: 1,000,000 CPU-hour core hours over 1 year.
  Application: Investigation of X-ray Emission from Interstellar Bow Shocks
  Technical Skills: Utilized hybrid MPI/OpenMP parallelisation, implemented job scheduling through SLURM for efficient resource allocation.
- Meluxina @ LuxProvide Data Center: Meluxina has 2 sockets per node, with 64 cores per socket and 2 CPUs per core
  Total Allocation: 500,000 CPU-hour core hours over 1 year.
  Application: Investigation of Wind-Blown Bow Shocks with NEMO

Teaching Experience

Indian Institute of Technology Guwahati Guwahati, India
Undergraduate Teaching Assistantship 2015-2019

- PH307 *Statistical Mechanics - Classical and Quantum* Aug-Nov 2015
- PH408, PH305 *Numerical Laboratory* July-Dec 2017
- PH412 *General Physics Laboratory I* Jan-May 2016
- PH511 *General Physics Laboratory II* July-Dec 2016 & July-Dec 2015
- PH001 *Preparatory Physics I - Newtonian Mechanics* Aug-Nov 2018 & Aug-Nov 2019
- PH002 *Preparatory Physics II - Electricity and Magnetism* First 2 weeks of Jan 2017 & Jan-April 2018
- PH002 *Preparatory Physics II - Electricity and Magnetism* Jan-April 2019

- Mid- and End-Semester *Examination related duties* All four semesters of 2017 & 2018

Darsana Academy Medical/Engineering Entrance Coaching Center Kottayam, Kerala, India
Instructor – *Senior Higher Secondary Physics* 6 Months, 2014

Marthoma College Thiruvalla, Kerala, India
Graduate Teaching (Unofficial) 2011-2012

- *Tensor Analysis (Mathematical Physics)* Sep 2011
- *Statistical Mechanics and Astrophysics* May 2012
- *Quantum Mechanics II* Nov 2012

Research Mentoring

- *Maximal mass of the neutron star with deconfined quark-gluon plasma core*
  PhD Project, Indian Institute of Technology Guwahati 2020
- *Reheating in scalar-tensor model with non-minimal coupling to curvature and non-minimal kinetic coupling of scalar field to the Einstein tensor*
  PhD Project, Indian Institute of Technology Guwahati 2020
- *Neutron star in $f(R,T)$ gravity with gravity-matter coupling*
  PhD Project, Indian Institute of Technology Guwahati 2020
- *Analysis of Narrow and Broad resonance regimes in the primordial reheating phase of the universe*
  Master's Projects, Indian Institute of Technology Guwahati April 2021
- *Scalar-tensor gravity and white dwarfs in Jordan frame*
  Master's Projects, Indian Institute of Technology Guwahati April 2019
- *Stability of compact stars with Harrison-Wheeler equation of state in the framework of general relativistic and Newtonian gravity*
  Master's Projects, Indian Institute of Technology Guwahati April 2018
- *Stability of White Dwarfs in General Relativity and Newtonian Gravity*
  Master's Projects, Indian Institute of Technology Guwahati April 2018
- *Analysis of Rotation curve of ESO138-G014 Galaxy in the framework of Modified Newtonian dynamics*
  B.Tech. Projects, Indian Institute of Technology Guwahati May 2020
- *Magnetic Field of Neutron Stars*
  B.Tech. Projects, Indian Institute of Technology Guwahati May 2020

Invited Talks and Presentations

- High Energy Physics Experiments and Cosmological Observations
  Amalambika School, Thekkady, India. Audience: High School Students 15 Dec 2009
- Safety Measures in the Kudankulam Nuclear Power Plant
  Marthoma College, Thiruvalla, India. Audience: General Public 10 Dec 2012
- Einstein and the Michelson-Morley Experiment
  Marthoma College, Thiruvalla, India Audience: General Public 16 Feb 2012
- Multi-ion Chemical Kinetics Modeling interstellar shockwaves and synthetic Emission Maps
  Instituto de Astronomía, UNAM, Mexico. Audience: Astrophysicists and Astronomers 12 Dec 2023
- General Relativistic Corrections to a White Dwarf Star
  Marthoma College, Thiruvalla, India Audience: Undergraduate and Master's Students 5 March 2013

- Stellar Structure of Quark Stars in Modified Starobinsky Gravity
  IITG, India. Audience: Undergraduate and Master's Students — 29 Nov 2019
- Effect of Quantum Spacetime Fluctuations on the Stability of White Dwarfs
  IITG, India. Audience: Undergraduate and Master's Students — 28 Nov 2018
- Quantum Cosmology with Generalized Uncertainty Relation
  IITG, India. Audience: Undergraduate and Master's Students — 2 Dec 2016

Skills and Tools

Programming	C++, C#, Python, git
CFD Developer	`NEMO` for `PION`,
Python Packages Developer	`NebulaPy`
Multiprocessing	Implemented Multiprocessing in NebulaPy
High-Performance Computing	Kay ICHEC, Meluxina LuxProvide Data Center
Spectral Synthesis Code	`Cloudy`, `ChiantiPy`, `PyNeb`
Documentation Generators	Sphinx, Doxygen
Utility Packages	Mathematica, Xmgrace, VisIt
Operating System	Linux/Unix, Mac OS
Communication	Excellent

Conferences

- Advances in Cool Evolved Stars
  Monash University Melbourne, Australia — 8th-12th July 2024
- Stellar Feedback in the ISM, IAU, Conference Summary: arXiv
  Instituto de Astronomía, UNAM, Mexico — 11-14 Dec 2023
- Irish National Astronomy Meeting 2023
  School of Physics, University College Cork, Ireland — 24-25 Sep 2023
- Irish National Astronomy Meeting 2022
  DIAS, Dunsink Observatory, Ireland — 1-2 Sep 2022
- Irish National Astronomy Meeting 2021
  Virtual — 1-3 Sep 2021
- Testing General Relativity using Gravitational Waves
  Virtual workshop organized jointly by the IACS and IITGN — 13-14 Aug 2020

Selected Outreach

- Hamilton Walk — 16 Oct 2024
- Festival of History at DIAS Dunsink Observatory — 10 Oct 2024
- Visitor Night at DIAS Dunsink Observatory — 25 Jan 2024
- Visitor Night at DIAS Dunsink Observatory — 2 Nov 2023
- Public Visitor Night at DIAS Dunsink Observatory — 19 Oct 2023
- Open House Dublin at DIAS Dunsink Observatory — 14 Oct 2023
- Culture Night, Dunsink Observatory — 22 Sep 2023

Other previous projects

– Expertise in $f(R, \phi)$ gravity.
– Surveyed Dark Matter evidence at various scales.
– Explored Scalar Field Dark Matter in scalar-tensor gravity.
– Verified cold boson dark matter halo formation.
– Investigated spin waves in magnetic lattices.